

INFRASTRUCTURE CAPEX & FUNCTIONAL ANALYSIS

Line-Item Cost Breakdown Framework

A Teachable Financial Ledger & Engineering Logic Deployment

Master Capital Allocation Overview

To secure a transparent, highly bankable investor framework, every single Rand spent must tie back directly to a specific biological defense, energy mitigation, or logistical requirement.

Asset Category Component	Primary Budget Share	Core Risk Eliminated
Structural Shells & Civil Foundations	40% of Total CAPEX	Climatic Exposure & Predation
Bio-Solar Energy & Co-Generation Microgrid	35% of Total CAPEX	Grid Deficits & Thermal Shock Mortalities
Hydrological, Biosecurity, & Automation Lifelines	25% of Total CAPEX	Pathogen Outbreaks & Operational Inefficiencies

Item 1: 384 m² Poultry Sprint Barn

Structural Allocation

Est. Budget: ZAR 320,000

- ✓ **Engineering Function:** High-density galvanized steel framework built to withstand highveld wind loads while optimizing natural interior cross-ventilation.
- ✓ **Biological Function:** Houses 5,760 broilers safely at an exact stocking density of 15 birds per square meter, preventing stress or overcrowding.



Stewardship Focus

The open-sided design utilizes zero-watt passive airflow, avoiding the heavy induction power traps of large tunnel ventilation fans.

Item 2: Winch PVC Curtains & Dwarf Wall

Climate Control Layout

Est. Budget: ZAR 45,000

- ✓ **Engineering Function:** Heavy-duty 550g/m² UV-stabilized PVC curtains combined with a 600mm plastered masonry brick wall.
- ✓ **Biological Function:** Dwarf wall stops floor-level freezing drafts during brooding while curtains seal completely within 90 seconds during highveld downpours.



Rain Defense

Prevents rainwater from entering, keeping floor litters dry to completely stop coccidiosis and respiratory outbreaks.

Item 3: 50mm Roof Insulation (PIR/EPS)

Infrastructure Entry	CAPEX Cost	Technical Function	Energy Loop Benefit
50mm High-Density Roof Insulation Boards	ZAR 65,000	Eliminates radiant solar heat gain in summer and seals thermal energy in winter.	Collapses active brooding heating energy demands by 35%, protecting BESS storage.

Item 4: 4-Pod Swine Nursery Block

The Isolation Fortress

Est. Budget: ZAR 140,000

- ✓ **Engineering Function:** Standalone 48 m² masonry shell divided by 150mm thick internal firewall barriers into four separate 12 m² airtight rooms.
- ✓ **Biological Function:** Acts as an unyielding immunological barrier that structures maximum virus risk to a single room.



Risk Capping

Caps potential pathogen exposure down to just 25% of the total batch, securing the remaining 75% of your live capital asset.

Item 5: Polypropylene Slatted Flooring

Sub-Floor Hygiene System

Est. Budget: ZAR 55,000

- ✓ **Engineering Function:** Interlocking high-density modular plastic slats elevated over a 400mm deep concrete flush gutter channel.
- ✓ **Biological Function:** Keeps piglets off cold concrete slabs to stop hypothermia. Manure drops down immediately to keep feet dry.



Pathogen Interruption

Separates animals from their waste instantly, stopping fecal-oral pathogen lifecycles without high water washing loads.

Item 6: 384 m² Swine Finisher Barn

Infrastructure Entry	CAPEX Cost	Structural Function	Biological Capacity
Open-Sided Finisher Barn (12m x 32m)	ZAR 280,000	Steel framework with 16 internal pens and concrete paths for easy sorting.	Houses 160 market hogs comfortably at a generous 1.2 m ² per head for deep grow-out.

Item 7: 30 kW / 30 kWh Off-Grid BESS

The Sovereign Power Core

Est. Budget: ZAR 420,000

- ✓ **Engineering Function:** High-capacity lithium iron phosphate (LFP) storage coupled to smart hybrid industrial inverters and rooftop solar arrays.
- ✓ **Biological Function:** Maintains continuous power for automated nursery floor heating loops and poultry brooding zones.



Grid Isolation

Insulates the farm from national energy grid dropouts, eliminating mass mortality risks from winter cold snaps.

Item 8: Biogas Digester & GenSet Loop

Co-Generation Engine

Est. Budget: ZAR 185,000

- ✓ **Engineering Function:** Anaerobic digester tank linked to gas scrubbers, storage bags, and a modified internal combustion generator set.
- ✓ **Biological Function:** Converts raw livestock waste into fuel, producing night volts to charge the BESS and steam heat for the nursery floors.



Closed Thermal Loop

Captures generator exhaust heat as steam, routing it sub-surface under nursery floors to cut electrical heating draws to zero.

Item 9: Hydrological Pumping & Towers

Infrastructure Entry	CAPEX Cost	Hydraulic Engineering Function	Asset Protection Limit
Solar Borehole Pump & Elevated Water Towers	ZAR 95,000	Low-induction DC pumps charge a high tower network using daytime solar power.	Holds a 5-day emergency water buffer to guarantee hydration survival during pump maintenance.

Item 10: UV & Inline Chlorination Dosing

Hydrological Sanitization Shield

Est. Budget: ZAR 38,000

- ✓ **Engineering Function:** Inline 50-micron sediment filter discs coupled to ultra-violet (UV) sterilizing tubes and kinetic chlorine dispensers.
- ✓ **Biological Function:** Neutralizes waterborne bacterial colonies and viruses before the water ever reaches internal nipple lines.



Biofilm Defeat

Kinetic chlorine injection eliminates organic biofilm inside drinker lines, stopping sub-clinical gut infections.

Item 11: Biosecurity Change-House Block

Human Access Firewall

Est. Budget: ZAR 75,000

- ✓ **Engineering Function:** A physical change-house layout separating outside dirty zones from inside clean production areas.
- ✓ **Biological Function:** Forces a strict shower-in and change protocol for all personnel, blocking human clothes and shoes from tracking external viruses onto the site.



Perimeter Gate

Includes a vehicle spray-dip boot at the primary gate line to neutralize tracking on vehicle delivery wheels.

Item 12: LoRaWAN IoT Sensor Nervous System

Infrastructure Entry	CAPEX Cost	Automation Data Function	Steward Alerts Generated
LoRaWAN IoT Sensor Array & Gateway	ZAR 42,000	Deploys continuous low-power radio tracking sensors for temperature, ammonia gas, and water flow.	Flags a 10% drop in water consumption or spikes in ammonia levels onto the main smart dashboard immediately.

FINANCIAL LEDGER ARCHIVED

Every Cost Justified. Every Asset Shielded.

Lwandile Engineering Projects (Pty) Ltd

Precision Alliance Budget Model | Just Energy Transition Submission